

Introduction

Sensitivity analyses complement the primary analysis by varying key assumptions – missing-data mechanism, model form, inclusion criteria – to assess how robust conclusions are. Tipping-point analyses identify how extreme assumptions must be before the primary conclusion flips, providing a concrete interpretation.

Prerequisites

Primary analysis; missing-data mechanisms; multiple imputation.

Theory

Tipping-point analysis under MI: impute missing outcomes in the experimental arm with an increasing shift δ (less favourable); re-pool. The smallest δ that makes the treatment effect no longer significant is the “tipping point”. A large tipping point means the conclusion is robust.

Other sensitivity analyses: different imputation methods, PP vs ITT, different covariate adjustments, varying inclusion criteria, alternative parametric models.

Assumptions

Sensitivity analyses are pre-specified; tipping points are interpreted clinically, not mechanically.

R Implementation

```
library(mice)

set.seed(2026)
n <- 200
arm <- factor(rep(c("ctrl", "trt"), each = n/2))
baseline <- rnorm(n, 5, 1)
outcome <- 0.6 * baseline + ifelse(arm == "trt", 0.8, 0) +
  rnorm(n, 0, 1)
outcome[sample(n, 40)] <- NA

df <- data.frame(arm, baseline, outcome)

# Base analysis under MAR
imp <- mice(df, m = 20, method = "pmm", printFlag = FALSE)
summary(pool(with(imp, lm(outcome ~ arm + baseline))))$estimate[2]

# Tipping-point analysis: penalise imputed trt-arm outcomes by delta
deltas <- seq(0, 2, by = 0.25)
effs <- sapply(deltas, function(d) {
  imp2 <- imp
  for (k in 1:20) {
    idx <- which(is.na(df$outcome) & df$arm == "trt")
    imp2$imp$outcome[[k]][df$arm[idx] == "trt"] <-
      imp2$imp$outcome[[k]][df$arm[idx] == "trt"] - d
  }
  summary(pool(with(imp2, lm(outcome ~ arm + baseline))))$estimate[2]
})
data.frame(delta = deltas, trt_effect = round(effs, 3))
```

Output & Results

Treatment effect declines linearly with δ ; the tipping point is where the effect crosses zero (or loses significance).

Interpretation

“A delta shift of 1.6 on the imputed intervention-arm outcomes was required to eliminate significance; clinically this would require intervention dropouts to fare 1.6 SD worse than MAR predicts. The primary conclusion is robust to plausible MNAR mechanisms.”

Practical Tips

- Pre-specify all sensitivity analyses in the SAP; post-hoc analyses are exploratory.
- Tipping points are most informative when expressed on the clinical scale.
- ICH E9 R1 promotes sensitivity-analysis thinking tied to the estimand.
- Running many sensitivity analyses is fine; interpret them holistically.
- Reporting a tipping-point figure in the primary paper is increasingly standard for high-missingness trials.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Diagnostic testing: Se, Sp, PPV, NPV, LR
- Kappa, ICC, Bland–Altman
- Biomarker statistics (Youden, NRI, decision curves)
- TRIPOD-AI, fairness auditing, reproducibility at scale